

- 1) A function's single most important role is to
- a. give a name to a block of code.
 - b. reduce program size.
 - c. accept arguments and provide a return value.
 - d. help organize a program into conceptual units.
- Ans: d. help organize a program into conceptual units.

2) A function itself is called the function definition.

- 3) Write a function called foo() that displays the word foo.
- Ans: void foo() {

```
    cout << "foo";  
}
```

4) A one-statement description of a function is referred to as a function declaration or a prototype.

5) The statement that carry out the work of the function constitute the function body.

6) A program statement that invokes a function is a function call.

7) The first line of a function definition is referred to as the declarator.

8) A function argument is

a) a variable in the function that receives a value from the calling program.

b) a way that functions resist accepting the calling program's values

c) a value sent to the function by the calling program.

↳ a value returned by the function to the calling program

Ans: C) A value sent to the function by the calling program.

9) True or false: When arguments are passed by value, the function works with the original arguments in the calling program.

Ans: False

10) What is the purpose of using argument names in a function declaration?

Ans: The purpose of using argument names are used to clarify the purpose of the argument.

11) Which of the following can legitimately be passed to a function?

a) A constant b) A variable c) A structure d) A header file.

Ans:

12) What is the significance of empty parentheses in a function declaration?

Ans: Empty parentheses means there is no argument taken by function.

13) How many values can be returned from a function?

Ans: There are only one values can be returned from a function.

14) True or false: When a function returns a value, the entire function call can appear on the right side of the equal sign and be assigned to another variable.

Ans: True

15. Where is a function's return type specified?

Ans: At the beginning of the declaration and declarator is a function's return type specified.

16. A function that does not return anything has return type void.

17. Here's a function:

```
int time2(int a)
```

```
{
```

```
    return (a * 2);
```

```
}
```

Write a main() program that includes everything necessary to call this function.

Ans:

```
#include <iostream>
```

```
using namespace std;
```

```
int time2(int a)
```

```
{
```

```
    return (a * 2);
```

```
}
```

```
int main() {
```

```
    int val = 5;
```

```
    int result;
```

```
time2
```

```
    result = time2(val);
```

```
    cout << "result: " << result << endl;
```

```
    return 0;
```

```
}
```

- 18) when an argument is passed by reference
- a. a variable is created in the function to hold the argument's value.
 - b. the function cannot access the argument's value
 - c. a temporary variable is created in the calling program to hold the argument's value.
 - d. the function accesses the argument's ~~original~~ original value in the calling program.

Ans: d. the function accesses the argument's original value in the calling program.

19) What is ~~the~~ a principal reason for passing arguments by reference

Ans: A principal reason for passing arguments by reference is modify the original argument.

20) Overloaded functions

- a. are a group of functions with the same name.
- b. all have the same number and types of arguments.
- c. make life simpler for programmers.
- d. may fail unexpectedly due to stress.

Ans: a. are a group of functions with the same name.

21) write declarations for two overloaded functions name bar(). They both return type int. The first take one argument of type char, and the second takes two arguments of type char. If this is impossible say why.

Ans:
→ `int bar(char);`
`int bar(char, char);`

22) In general, an inline function executes faster than a normal function, but requires more memory.

23) write the declarator for an inline function named foobar() that takes one argument of type float and returns type float.

Ans →

inline float foobar(float fval);

24) A default argument has a value that
a. may be supplied by the calling program.

b. may be supplied by the function

c. must have a constant value.

d. must have a variable value.

Ans: a. may be supplied by the calling program.

25) write a declaration for a function called blyth() that takes two arguments and returns type char. the first argument type is int, and the second type float with a default value of 3.14159.

Ans:

char blyth(int, float val=3.14159);

26) Scope and storage class are concerned with the visibility and lifetime of a variable.

27) What functions can access a global variable that appears in the same file with them?

Ans: those functions defined following the variable definition.

28) What functions can access a local variable?

Ans: A function can access a local variable which is defined in function.

29) A static local variable is used to

- a. make a variable visible to several functions.
- b. make a variable visible to only one function.
- c. conserve memory when a function is not executing.
- d. retain a value when a function is not executing.

Ans: b. make a variable visible to only one function.

30) In what unusual place can you use a function call when a function returns a value by reference?

Ans: On the left side of the equal sign ~~unusual~~ unusual place can we use a function call when a function returns a value by reference.